

Machine Learning

Week -4

Machine Learning

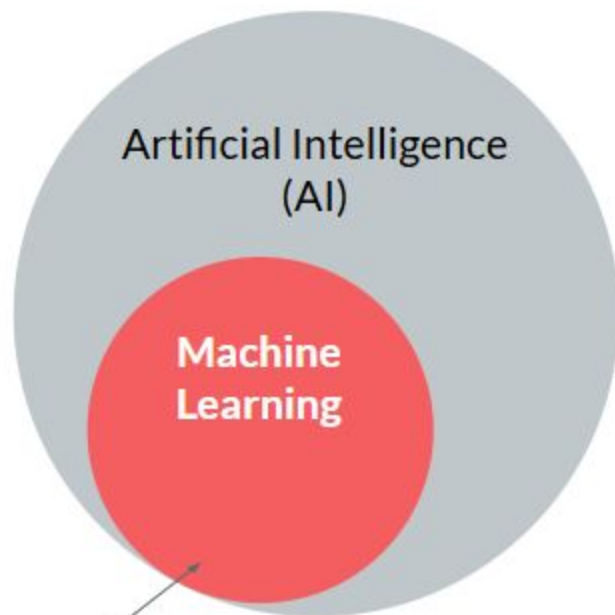
- The term **Machine Learning** was coined by **Arthur Samuel** in 1959:

“Machine Learning algorithms enable the computers to **learn from data**, and even **improve** themselves, **without being explicitly programmed**”.

- In 1997, Tom Mitchell gave a mathematical and relational definition that:

“A computer program is said to learn from **experience E** with respect to some **task T** and some **performance measure P**, if its performance (as measured by P) on T improves with experience E”.

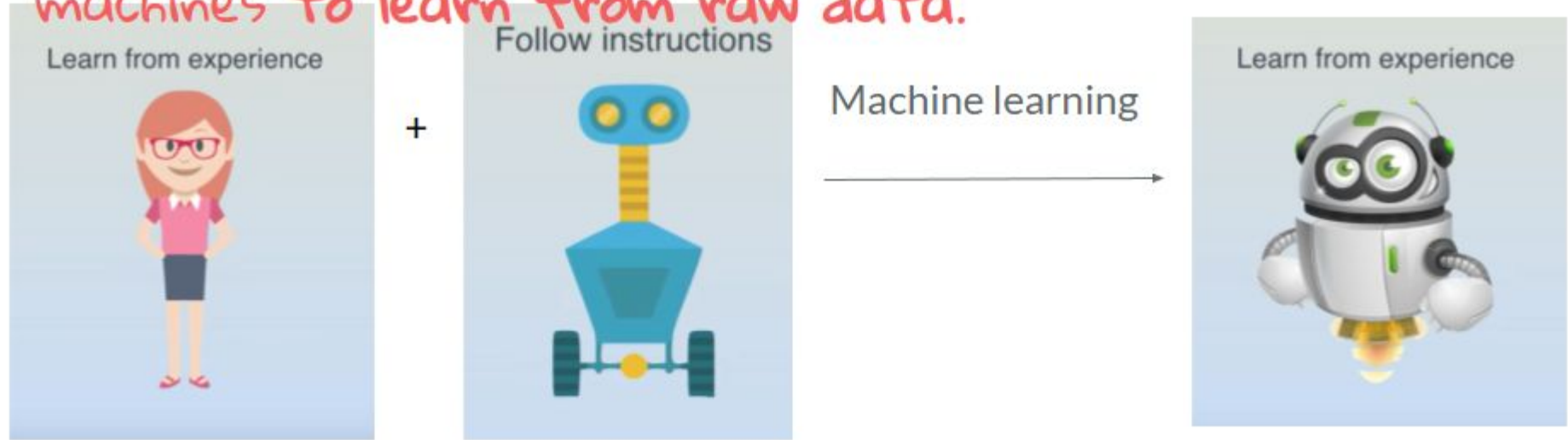
What is machine learning?



We call something **machine learning** when instead of telling a computer to do something, we allow a computer to come up with its own solution based upon the data it is given.

Using data science methods and sometimes big data

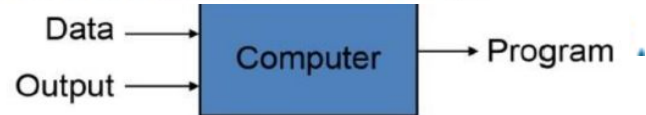
Machine learning is a subset of AI that allows machines to learn from raw data.



Humans learn from experience. Traditional software programming involves giving machines instructions which they perform. **Machine learning involves allowing machines to learn from raw data so that the computer program can change when exposed to new data (learning from experience).**



Traditional Programming



Machine Learning

learning

- There

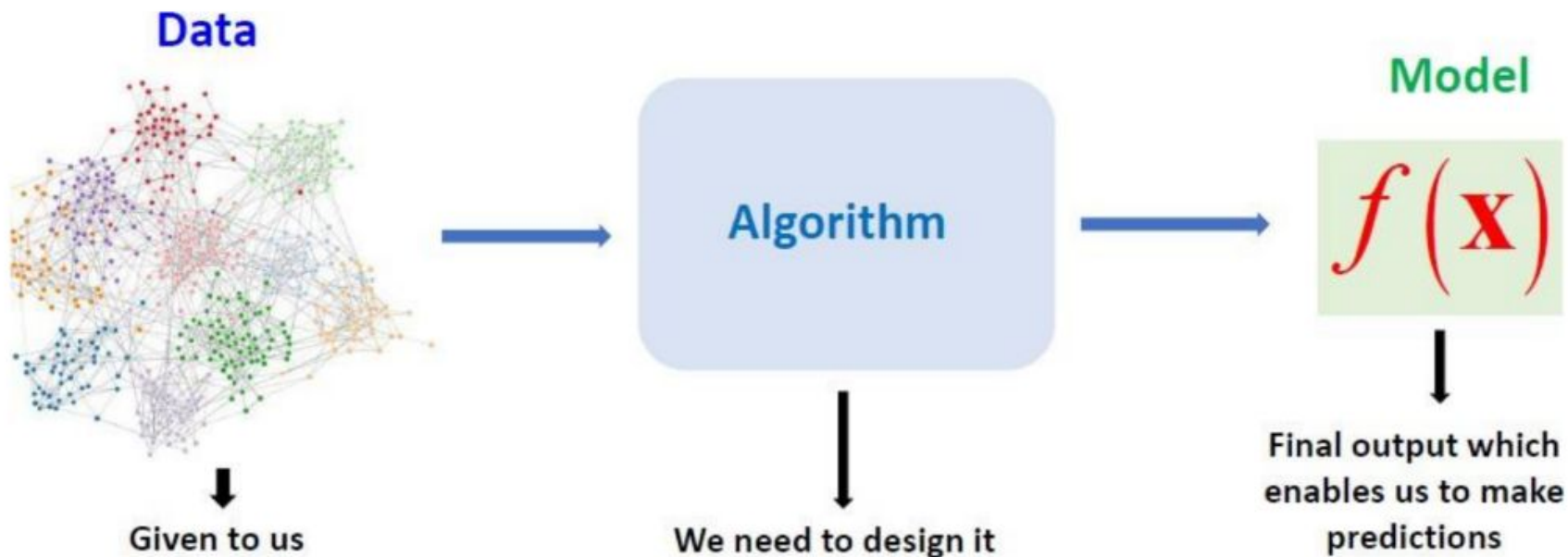
- There are huge amounts of data generated every day.
- Previously impossible problems are now solvable.
- Companies are increasingly demanding quantitative solutions.

“Every day, we create 2.5 quintillion bytes of data – so much that 90% of the data in the world today has been created in the last two years alone.” [1]



What is Machine Learning?

- Given examples (training data), make a machine learn system behavior or discover patterns



Machine learning example: Predicting malaria



Dr. Delta works to diagnose patients with malaria.

However, it takes a long time for her to see everyone.



Luckily, Dr. Delta has **historical patient data about what factors predict malaria**, such as body temperature, travel history, age, medical history.



Dr. Delta can use historical data as an input in a **machine learning algorithm** to help her predict whether a new patient will have malaria.

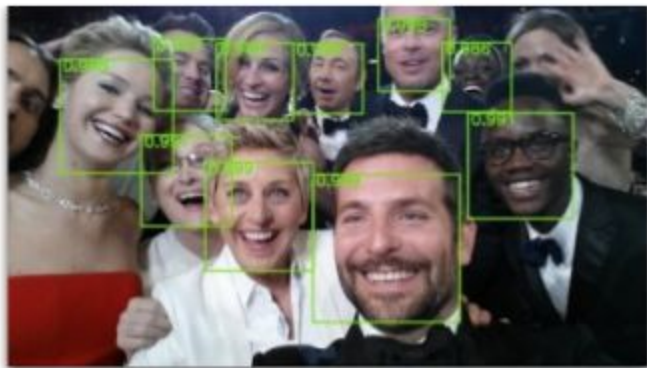
The algorithm (the machine) learns from past data, like a human would, and is thus able to make predictions about the future.



Machine learning is a powerful tool; it can...



Determine your credit rating based upon cell phone usage.

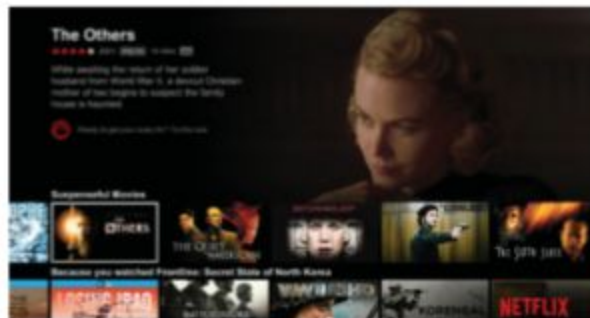


Recognize your face in a photo.

Emmanuel Macron Is Inaugurated as French President

Ceremony comes a week after victory over Marine Le Pen in presidential election.

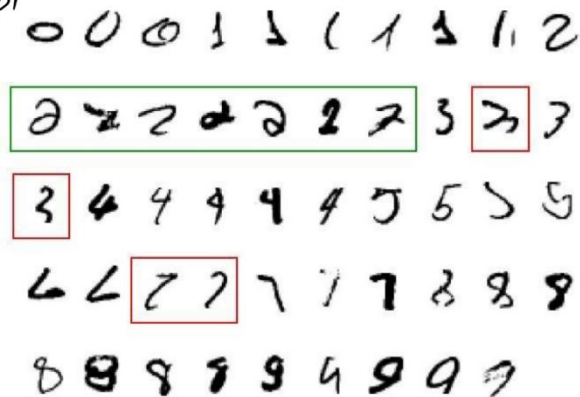
Determine the topic of a piece of text.



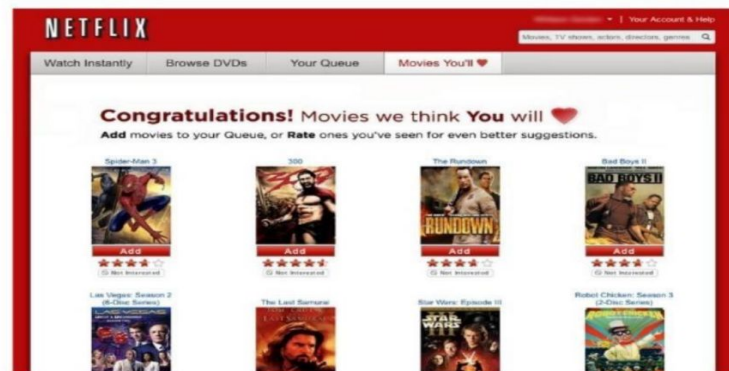
Recommend movies you will like.



Classical Example: Recognize hand-written ?!



Example: Netflix



Luxury hotels in Istanbul



Example 1:

- Suppose you decide to check out trip
- You browse through the travel agency website and search for a hotel
- When you look at a specific hotel, just below the hotel description there is a section titled “You might also like these hotels”.
- This is a common use case of Machine Learning called “Recommendation Engine”
 - Many data points were used to train a model in order to predict what will be the best hotels to show you under that section, based on a lot of information they already know about you

Machine learning (ML)

- Machine learning studies the design and development of algorithms that learn from the data and improve their performance through experience.
- ML refers to a set of methods and that help computers to learn, optimize and adapt on their own.
- ML has been employed to devise algorithms for diverse applications including:
 - object detection or identification in computer vision,
 - sentiment analysis of speaker or writer,
 - detection of disease and planning of therapy in healthcare,
 - product recommendation in e-commerce,
 - learning strategies for playing games,
 - fraudulent transaction detection or loan application approval in banking sector

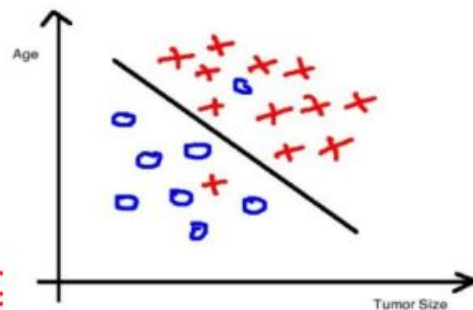
Applications of machine learning

- **Medical diagnoses:** ML is trained to recognize cancerous tissues
 - “Is this cancer?”
- **Graph Processing:**
 - “Which of these people are good friends with each other?”
- **Recommender Systems:**
 - “Will person X likes movie Y?” recommending movies to customers
- **Financial industry and trading** —fraud investigations and loan sanction
- **Speech Recognition:**
 - “Is this his/her voice?” (voice searches, voice dialing, call routing, and appliance control)

Such problems are excellent targets for Machine Learning, and in fact machine learning has been applied to such problems with great success.

What is a Model

- A **model** is a mathematical formula with a number of parameters that need to be learned from the data
 - Fitting a model to the data is a process known as **model training**
- **Example:** Consider a one feature/variable linear regression, where the goal is to fit a line (described by the equation $y = ax + b$) to a set of distributed data points.
- Once the model training is completed we get a model equation $y = 2x + 5$.
 - Then for a set of inputs [1, 0, 7, 2, ...] we would get a set of outputs [7, 5, 19, 9, ...].



What is a Model

A **model** is a simplified representation of a real-world system used to **understand, explain, or predict** something.

A model is a **mathematical equation or algorithm** that learns patterns from data and makes predictions.

Suppose you want to predict **student marks** based on **study hours**.

You collect data:

Hours Studied	Marks
1	40
2	50
3	60
4	70
5	80

From this, you notice a pattern:

More hours → Higher marks

A simple mathematical model could be:

$$Marks = 10 \times Hours + 30$$

This equation is your **model**.

Model =

 Input →  Mathematical Pattern →  Output

Example:

- Input: Hours studied
- Model: Equation
- Output: Predicted marks

$$y = mx + c$$

— **Alaorithm vs Model**

An **algorithm** is a **procedure** or **set of steps** used to solve a problem.

It tells the computer **how to learn** from data.

Example: Think of it as a *recipe*.

- Linear Regression algorithm
- Decision Tree algorithm
- K-Means algorithm

An algorithm **creates** a model.

Algorithm vs Model

A **model** is the **final learned output** produced by an algorithm after training on data.

It contains:

· Think of it as the *trained result* of the recipe.

- Learned parameters
- Mathematical relationship
- Pattern extracted from data

Algorithm = Learning Method
Model = Learned Knowledge

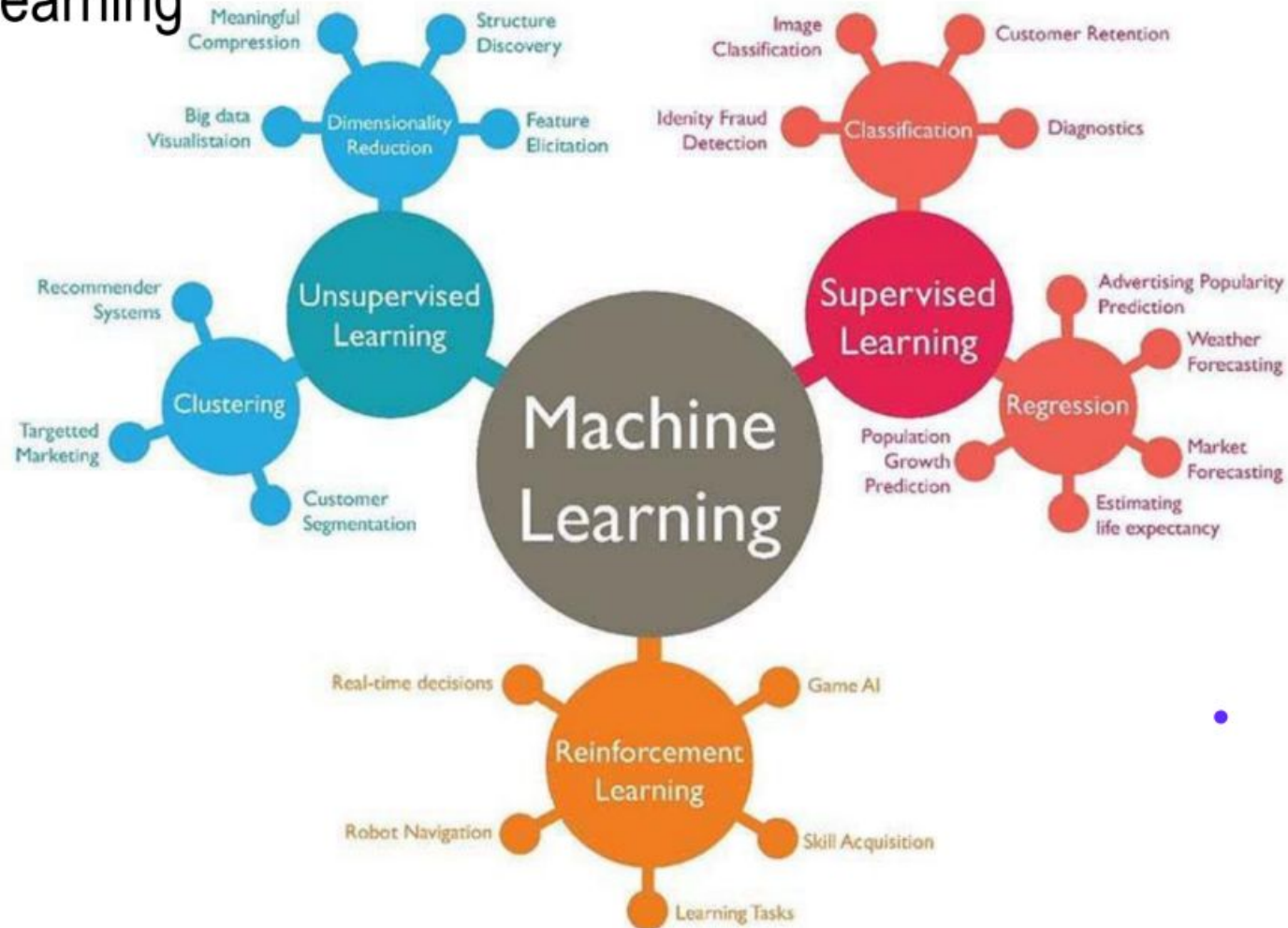
Concept	Real-Life Example
Algorithm	Cooking recipe
Model	Final cooked dish

Algorithm vs Model

- The linear regression algorithm produces a model, that is, a vector of values of the coefficients of the model (e.g. $y = 2x + 5$).
- The decision tree algorithm produces a model comprised of a tree of if-then statements with specific values.
- A neural network along with backpropagation + gradient descent: produces a model comprised of a trained (weights assigned) neural network.

Types of Machine Learning

Machine learning can be classified into 3 types of algorithms.



Types of Algorithm in ML

☐ Supervised learning .

- Prediction
- Classification (discrete labels), Regression (real values)

☐ Unsupervised learning.

- Clustering
- Probability distribution estimation
- Finding association (in features)
- Dimension reduction

☐ Semi-supervised learning.

☐ Reinforcement learning.

- Decision making (robot, chess machine)

1. Supervised Learning

The most common type of machine learning is Supervised Learning. In this approach, the algorithm is trained on a labelled dataset, i.e., each training example is paired with an output label. The objective is to learn to map inputs to the correct output, making it ideal for tasks such as classification and regression.

For example - a supervised learning algorithm can be trained to recognise images of cats and dogs if provided with numerous labelled images of both animals.

2. Unsupervised Learning

On the contrary to supervised learning, unsupervised learning deals with unlabelled data. The system tries to learn the underlying patterns and structures from the input data without explicit instructions on what to predict. Clustering and association are two primary types of unsupervised learning tasks. Clustering algorithms, such as K-means, group data points with similar characteristics.

This is specifically beneficial for market segmentation or finding anomalies in data.

3. Semi-Supervised Learning

Semi-supervised learning is a hybrid approach that combines a small amount of labelled data with a large amount of unlabeled data during training. This method is particularly useful when acquiring a fully labelled dataset is expensive or time-consuming.

Semi-supervised learning can significantly improve learning accuracy and is often used in natural language processing and medical image analysis.

4. Reinforcement Learning

Reinforcement learning is inspired by the ethos of behavioural psychology. The process involves training an agent to make a sequence of decisions by rewarding desired behaviours and penalising undesired ones. The agent learns by interacting with its environment, receiving feedback from its actions, and optimising its strategy to achieve the best outcomes.

This type of learning is extensively used in robotics, game development, and self-driving cars.

Supervised Learning

- A supervised learning algorithm takes a known set of input dataset and its known responses to the data (output) to learn the model.
- A learning algorithm then trains a model to generate a prediction for the response to new data or the test dataset.
- Supervised learning uses classification algorithms and regression techniques to develop predictive models.
- The algorithms include linear regression, logistic regression, neural networks, decision tree, Support Vector Machine (SVM), random forest, naive Bayes, and k-nearest neighbor.

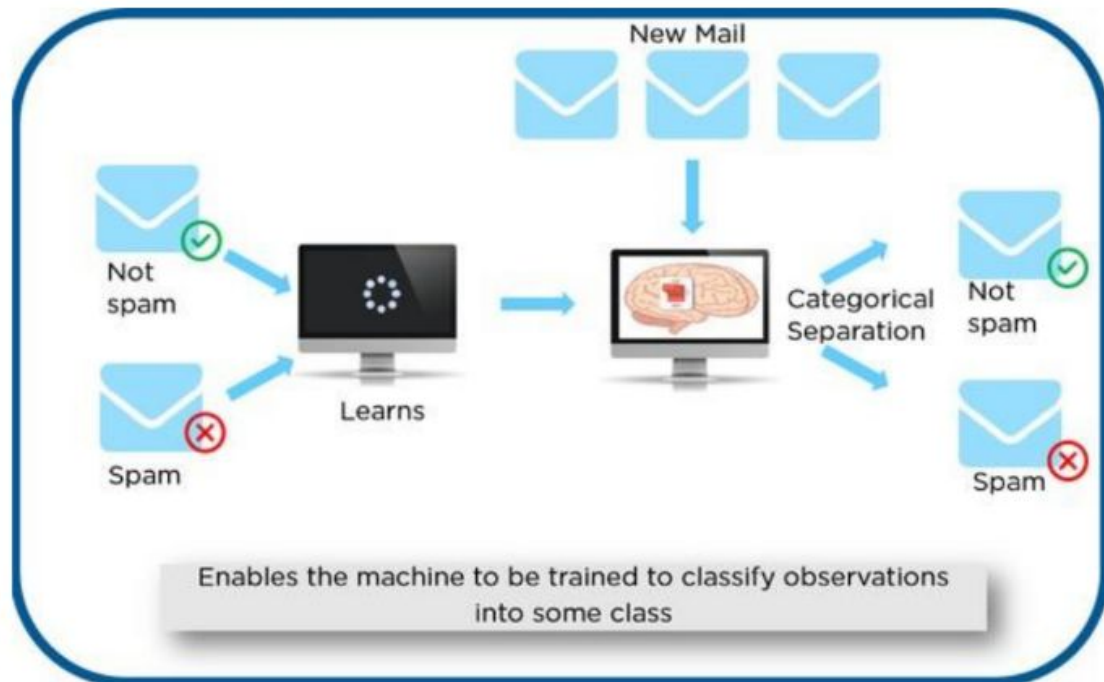
Supervised Learning

For example:

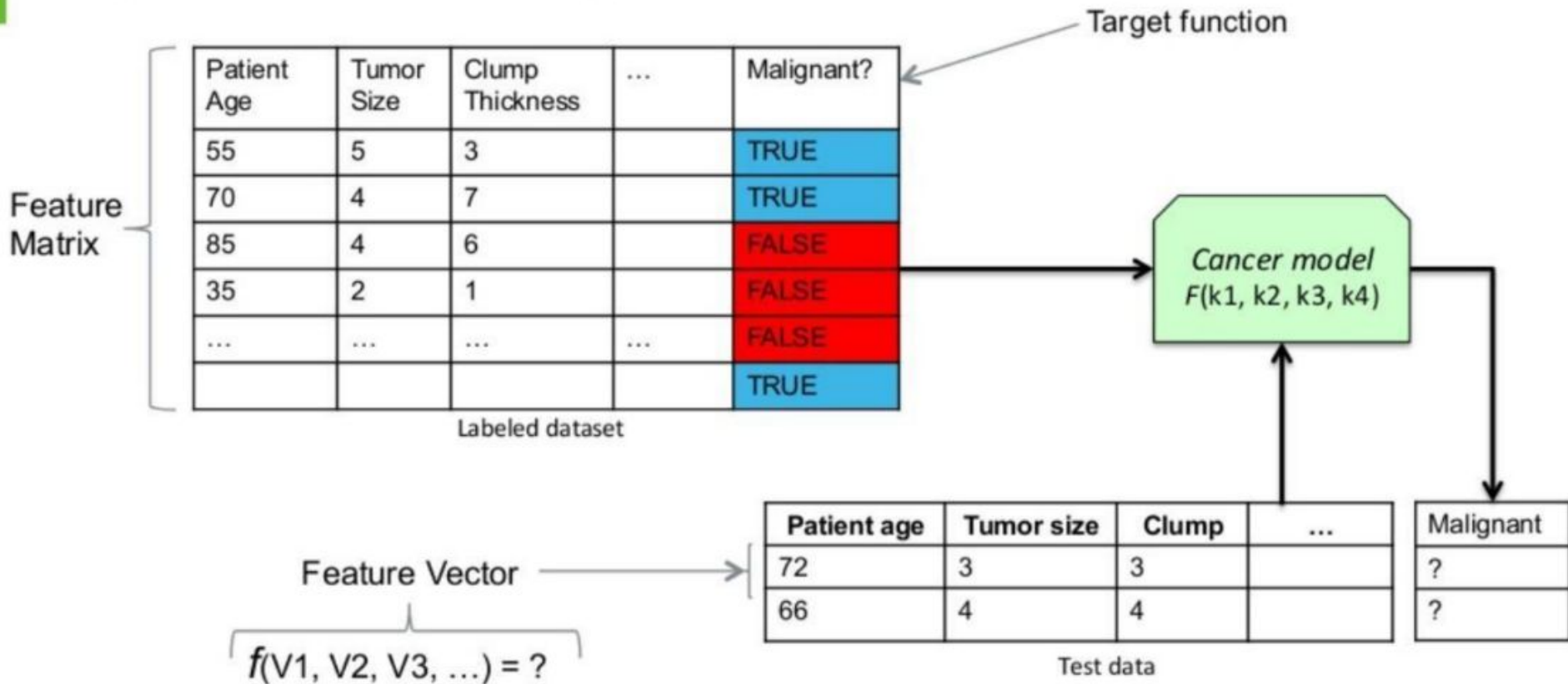
Spam filtering where large number of email messages are labelled as either:

- spam
- non-spam

- New email message will then be classified as spam or non-spam



Supervised Learning: learn from examples



Supervised Learning

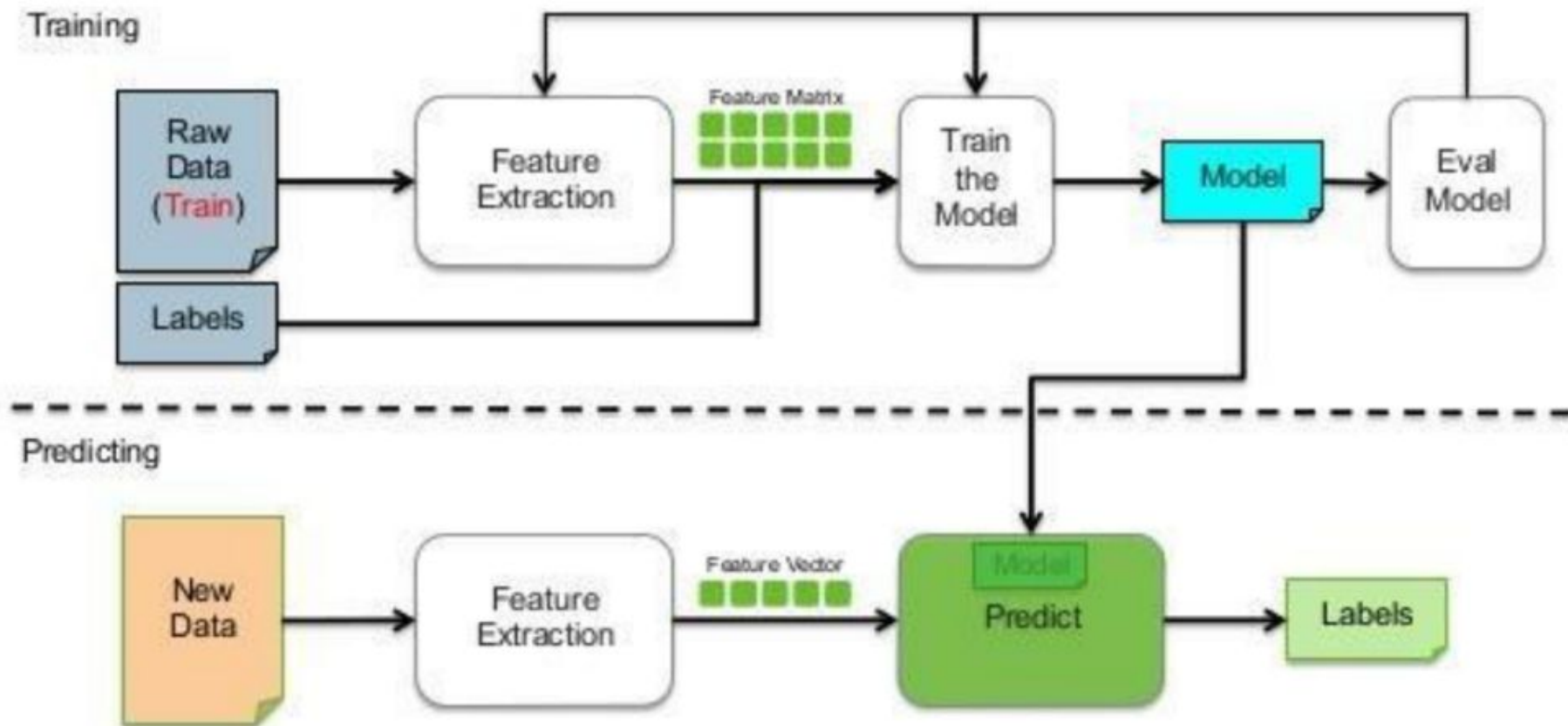


Training
Images



Test
Image

Supervised Learning Workflow



Supervised Learning

- **Regression algorithms** attempt to estimate the mapping function (f) from the input variables (x) to numerical or continuous output variables (y).

The model learns:

$$Y = f(X)$$

Types:

- **Regression** → Predict continuous values
Example: Temperature prediction, House price
- **Classification** → Predict categories
Example: Spam detection, Disease detection

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- SVM
- Neural Networks

Supervised Learning

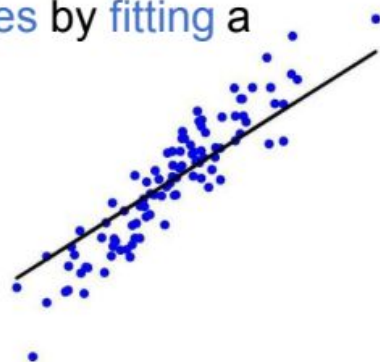
Regression

Regression: Quantitative Prediction on a continuous scale

A linear regression attempts to model the relationship between two variables by fitting a linear equation to observed data

Examples: Prediction of

- Age of a person from his/her photo
- Price of 10 Marla, 5-bedroom house in 2050
- USD/PKR exchange rate after one week
- Efficacy of Pfizer Covid vaccine
- Average temperature/Rainfall during the monsoon
- Cumulative score in ML course
- Probability of a decrease in the electricity prices in Pakistan
- No. of steps per day

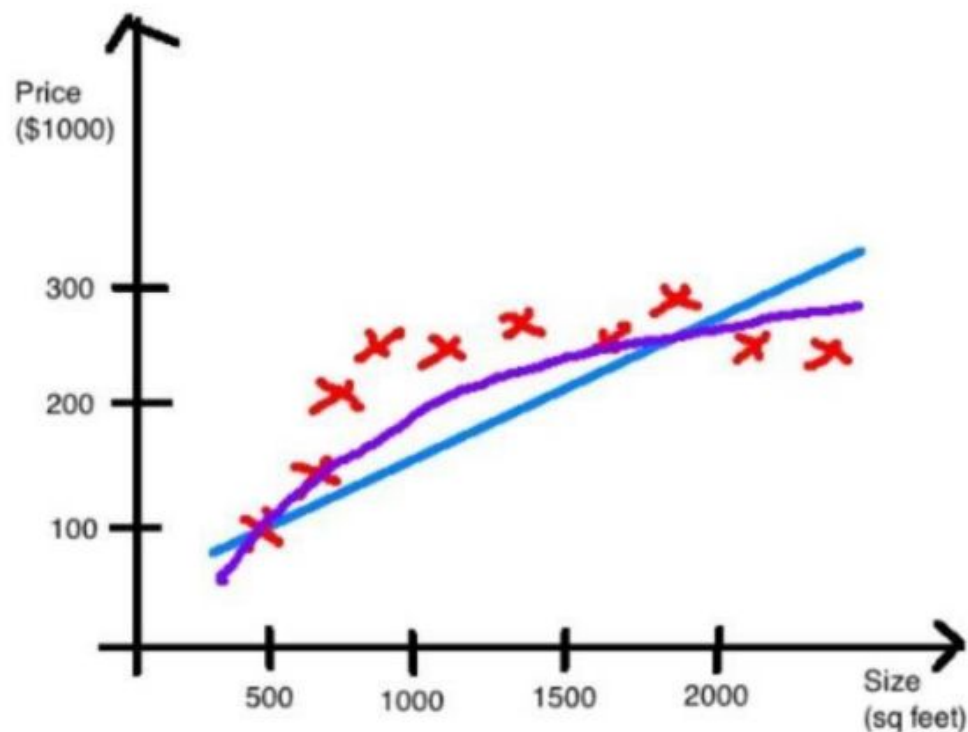


What do all these problems have in common?

Continuous outputs

Predicting continuous outputs is called **regression**

Regression: predict a continuous value

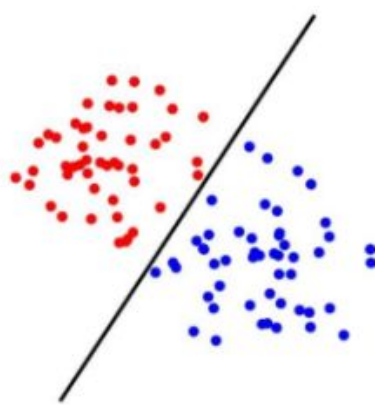


Some techniques:

- Linear Regression / GLM
- Decision Trees
- Support vector regression
- SGD
- Ensembles

Supervised Learning

Classification



- **Classification:** Given a data sample, predict its class (discrete)
- **Examples: Prediction of**
 - Gender of a person using his/her photo or hand-writing style
 - Spam filtering: To check whether an email is genuine or spam
 - Object or face detection in a photo
 - We will be back on Campus on Aug 22
 - Temperature/Rainfall normal or abnormal during monsoon
 - Letter grade in ML course
 - Decrease expected in electricity prices in Pakistan next year

What do all these problems have in common?

Discrete outputs:
Categorical Yes/No
(Binary Classification)
Multi-class
classification: multiple
classes

Predicting a categorical output is called **classification**

Classification

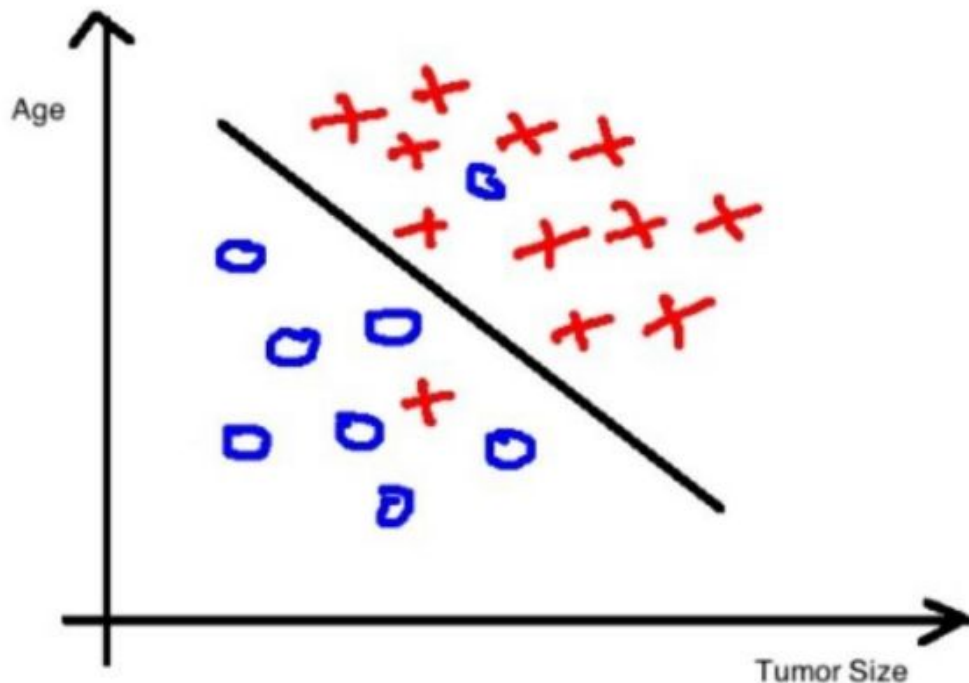
Classification learns from existing categorizations and then assigns unclassified items

- Classification models classify input data into categories and **predict discrete responses**
- Classification is recommended if the data can be **categorized, tagged, or separated into specific groups or classes**
- **Classification Examples:**
 - To recognize letters and numbers in Handwriting
 - To detect whether a tumor is benign or cancerous
- **Classification Algorithms:**
 - k-nn, Decision Trees, Random Forest, SVM, Neural Network...

Classification Algorithms

- **Classification algorithms** attempt to estimate the mapping function f from the input variables (x) to discrete or categorical output variables (y) .
 - In this case, y is a category that the mapping function predicts.
- **For example**, when provided with a dataset about houses, a classification algorithm predict whether the prices for the houses “sell more or less than the recommended retail price”
- **For example**, in a banking application, the customer who applies for a loan may be classified as a safe and risky according to his/her age and salary. The constructed model can be used to classify new data

Classification: predicting a category



Some techniques:

- Naïve Bayes
- Decision Tree
- Logistic Regression
- SGD
- Support Vector Machines
- Neural Network
- Ensembles

Supervised Learning Setup

Nomenclature

In these regression or classification problems, we have

- **Inputs**—referred to as Features
- **Output**—referred to as Label
- **Training data**—(input, output) for which the output is known and is used for training a model by ML algorithm
- **A Loss, an objective or a cost function**—determines how well a trained model approximates the training data
- **Test data**—(input, output) for which the output is known and is used for the evaluation of the performance of the trained model

Supervised Learning

Setup

Nomenclature -

Example

Predict Stock Index

Price

- Features (Input)
- Labels (Output)
- Training data

Interest_Rate	Unemployment_Rate	Stock_Index_Price
2.75	5.3	1464
2.5	5.3	1394
2.5	5.3	1357
2.5	5.3	1293
2.5	5.4	1256
2.5	5.6	1254
2.5	5.5	1234
2.25	5.5	1195
2.25	5.5	1159
2.25	5.6	1167
2	5.7	1130
2	5.9	1075
2	6	1047
1.75	5.9	965
1.75	5.8	943
1.75	6.1	958
1.75	6.2	971
1.75	6.1	949
1.75	6.1	884
1.75	6.1	866
1.75	5.9	876
1.75	6.2	?
1.75	6.2	?
1.75	6.1	?

Supervised Learning Setup

We assume that we have d columns (features) of the input. In this example, we have two features; interest rate and unemployment rate, that is, $d = 2$.

In general, we use $\mathbf{x}_i$ to refer to features of the i -th sample, that is,

$$\mathbf{x}_i = [x_{i,1}, x_{i,2}, x_{i,3}, \dots, x_{i,d}]$$

If y_i is the label associated with the i -th sample $\mathbf{x}_i$, we formulate training data in pairs as

$$(\mathbf{x}_i, y_i), \quad i = 1, 2, \dots, n$$

Here, n denotes the number of samples in the training data. In this example, we have $n = 21$

Interest_Rate	Unemployment_Rate	Stock_Index_Price
2.75	5.3	1464
2.5	5.3	1394
2.5	5.3	1357
2.5	5.3	1293
2.5	5.4	1256
2.5	5.6	1254
2.5	5.5	1234
2.25	5.5	1195
2.25	5.5	1159
2.25	5.6	1167
2	5.7	1130
2	5.9	1075
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1.75	6.1	949
1.75	6.1	884
1.75	6.1	866
1.75	5.9	876
1.75	6.2	?
1.75	6.2	?
1.75	6.1	?

Supervised Learning Setup

Formulation

Using the adopted notation, we can formalize the supervised machine learning setup. We represent the entire training data as

$$D = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)\} \subseteq \mathcal{X}^d \times \mathcal{Y}$$

Here $\mathcal{X}^d$ - d dimensional feature space and $\mathcal{Y}$ is the label space.

Regression:

$\mathcal{Y} = \mathbf{R}$ (prediction on continuous scale)

Classification:

$\mathcal{Y} = \{0, 1\}$ or $\mathcal{Y} = \{-1, 1\}$ or $\mathcal{Y} = \{1, 2\}$ (Binary classification)

$\mathcal{Y} = \{1, 2, \dots, M\}$ (M-class classification)

Supervised Learning Setup

Example

Data of 200

Patients:

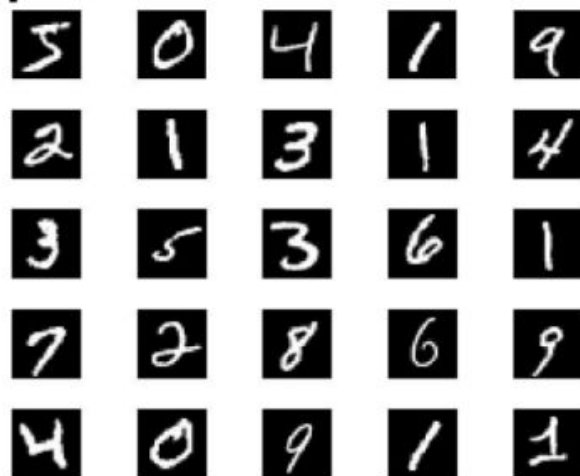
- Age of the patient
- Cholesterol levels
- Glucose levels
- BMI
- Height
- Heart Rate
- Calories intake
- No. of steps taken



Supervised Learning Setup

Example:

e:



Prediction

MNIST Data:

- Each sample 28x28 pixel image
- 60,000 training data
- 10,000 testing data

Supervised Learning Setup

Recall a problem in hand. We want to develop a model that can predict the label for the input for which label is unknown.

We assume that the data points $(\mathbf{x}_i, y_i)$ are drawn from some (unknown) distribution $P(X, Y)$.

Our goal is to learn the machine (model, function or hypothesis) h such that for a new pair $(\mathbf{x}, y) \sim P$, we can use h to obtain

$$h(\mathbf{x}) = y$$

with high probability or

$$h(\mathbf{x}) \approx y$$

in some optimal sense.

Supervised Learning Setup Hypothesis Class

We call the set of possible functions or candidate models (linear model, neural network, decision tree, etc.) “the hypothesis class”.

Denoted by $\mathcal{H}$

For a given problem, we wish to select hypothesis (machine) $h \in \mathcal{H}$.

Q: How?

A: Define hypothesis class $\mathcal{H}$ for a given learning problem.

Evaluate the performance of each candidate function and choose the best one.

Supervised Learning Setup

Q: How do we evaluate the performance?

A: Define a loss function to quantify the accuracy of the prediction.

Loss Function:

Loss function should quantify the error in predicting y using hypothesis function h and input $\mathbf{x}$.

Denoted by $\mathcal{L}$.

Supervised Learning Setup

0/1 Loss Function:

Zero-one loss is defined as:

$$\mathcal{L}_{0/1}(h) = \frac{1}{n} \sum_{i=1}^n 1 - \delta_{h(\mathbf{x}_i) - y_i}$$

Here $\delta_{h(\mathbf{x}_i) - y_i}$ is the delta function defined as

$$\delta_k = \begin{cases} 1, & k = 0 \\ 0 & \text{otherwise} \end{cases}$$

Interpretation:

- **Normalize the loss by the total number of training samples, n , so that the output can be interpreted as the average loss per sample.**
- **Loss function counts the number of mistakes made by hypothesis function D .**
- **Not used frequently due to non-differentiability and non-continuity.**

Supervised Learning Setup

Squared Loss Function:

Squared loss is defined as (also referred to as mean-**square error, MSE**)

$$\mathcal{L}_{\text{sq}}(h) = \frac{1}{n} \sum_{i=1}^n (h(\mathbf{x}_i) - y_i)^2$$

Interpretation:

- Again note normalization by the number of samples.
- Loss grows quadratically with the absolute error amount in each sample.

Root Mean Squared Error (RMSE):

RMSE is just square root

$$\mathcal{L}_{\text{rms}}(h) = \sqrt{\frac{1}{n} \sum_{i=1}^n (h(\mathbf{x}_i) - y_i)^2}$$

Supervised Learning Setup

Absolute Loss Function:

Absolute loss is defined as

$$\mathcal{L}_{\text{abs}}(h) = \frac{1}{n} \sum_{i=1}^n |h(\mathbf{x}_i) - y_i|$$

- **Interpretation:**

- Loss grows linearly with the absolute of the error in each prediction.
- Used in regression and suited for noisy data.

* All of the losses are non-negative

Supervised Learning Setup

To illustrate this, let us consider a model h trained on every input in D , that is, giving zero loss. Such function is referred to as memorizer and can be formulated as follows

$$h(\mathbf{x}) = \begin{cases} y_i, & \exists (\mathbf{x}_i, y_i) \in D, \quad \mathbf{x}_i = \mathbf{x}, \\ 0, & \text{otherwise} \end{cases}$$

Interpretation:

- 0% loss error on the training data (Model is fit to every data point in D).
- Large error for some input not in D
- First glimpse of **overfitting**.

Revisit:

Q: How can we ensure that hypothesis h will give low loss on the input not in D ?

A: Train/Test Split

Supervised Learning Setup

Generalization: The Train-Test Split

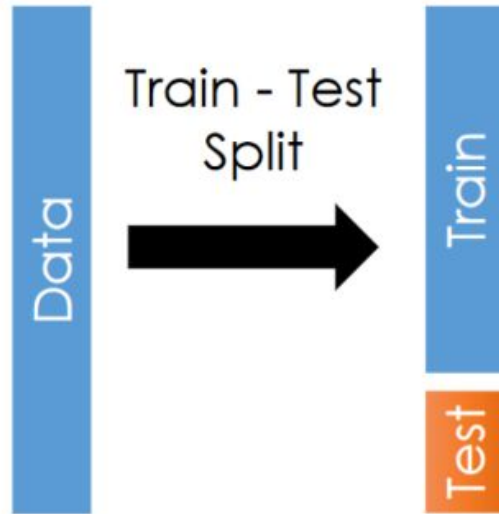
To resolve the overfitting issue, we usually split D into train and test subsets:

- D_{TR} as the training data, (70, 80 or 90%)
- D_{TE} as the test data, (30, 20, or 10%)

How to carry out splitting?

- Split should be capturing the variations in the distribution.
- Usually, we carry out splitting using i.i.d. sampling and time series with respect to time

You can only use the test dataset once after deciding on the model using training dataset



Supervised Learning Setup

Generalization: The Train-Test Split

At times, we usually split D into three subsets, that is, the training data is further divided into training and validation datasets:

- D_{TR} as the training data, (80%)
- D_{VA} as the validation data, (10%)
- D_{TE} as the test data, (10%)

Q: Idea:

Validation data is used to evaluate the loss for a function h that is determined using the learning on the training data-set. If the loss on validation data is high for a given h , the hypothesis or model needs to be changed.

Supervised Learning Setup

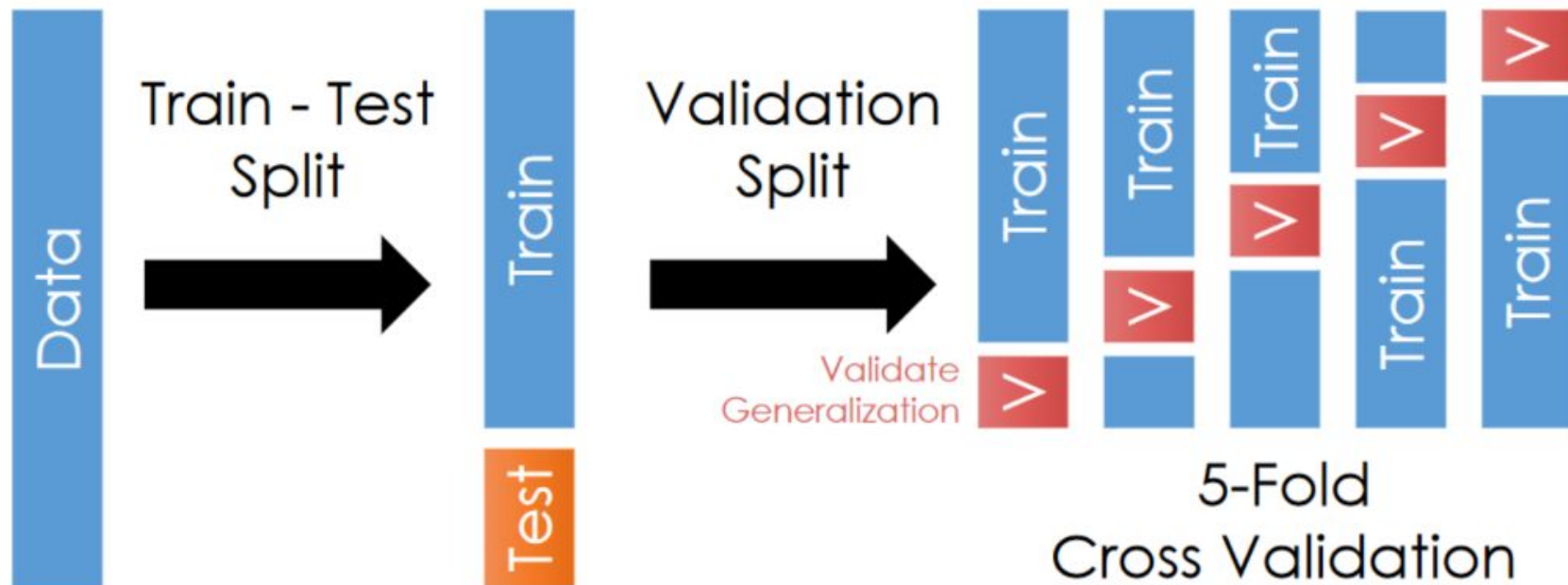
Generalization: The Train-Test Split

More explanation* to better understand the difference between validation and test data:

- **Training set:** A set of examples used for learning, that is to fit the parameters of the hypothesis (model).
- **Validation set:** A set of examples used to tune the hyper-parameters of the hypothesis function, for example to choose the number of hidden units in a neural network OR the order of polynomial approximating the data.
- **Test set:** A set of examples **used** only to assess the performance of a fully-specified model or hypothesis.

Supervised Learning Setup

Generalization: The Train-Test Split (Example)



Cross validation simulates multiple train-test splits on the training data

Unsupervised Learning

In unsupervised learning the training data comprises examples of input vectors WITHOUT any corresponding target variables.

- In unsupervised learning, the algorithm **learns from plain examples without any associated response**, leaving to the **algorithm to determine the data patterns on its own**.
 - In an **unsupervised learning** you only have input data (X) and no corresponding output variables.
- The goal for unsupervised learning is to model the **underlying structure** or **distribution in the data** in order to learn more about the data.

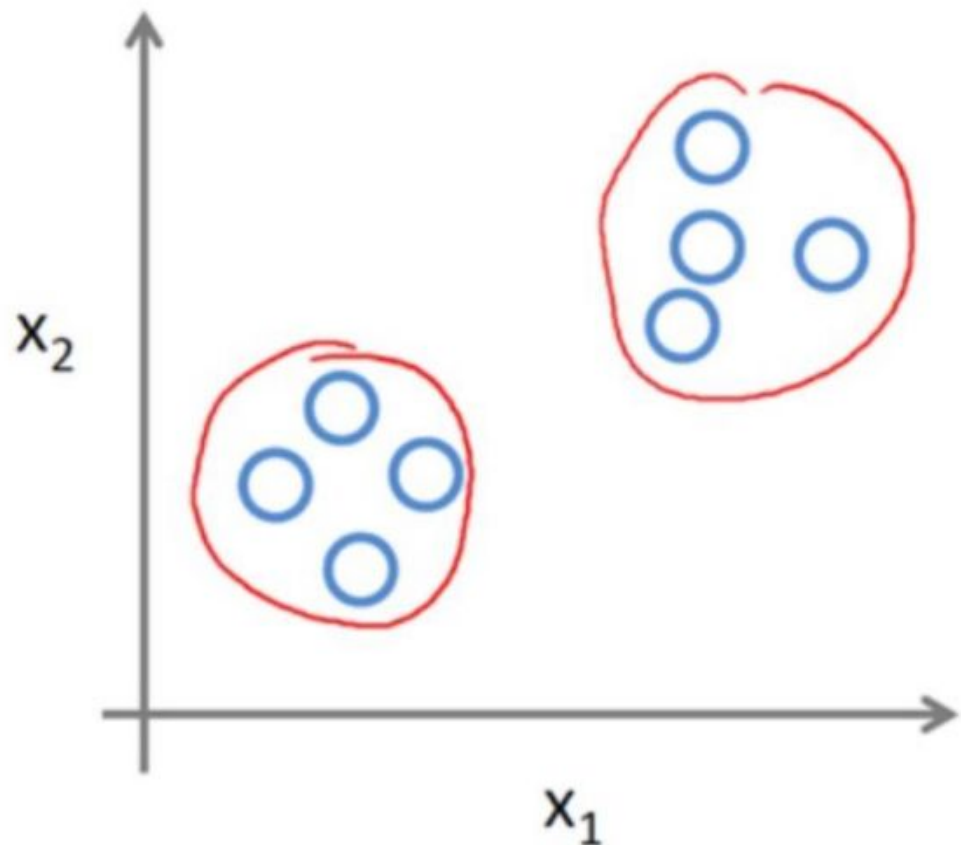


Unsupervised Learning

Examples: Speech recognition, document clustering, and image compression.

- In **document clustering**, **the aim** is to group documents into various reports of politics, entertainment, sports, culture, heritage, art, and so on.
- **Fraud Detection**: Identify groups of motor insurance policy holders with a high average claim cost
- **Social Networks**: Recognize communities within large groups of people

Clustering: detect similar instance groupings



Some techniques:

- k-means
- Spectral clustering
- DB-scan
- Hierarchical clustering

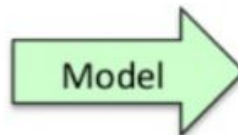
Unsupervised Learning: detect natural patterns

Clip slide

Age	State	Annual Income	Marital status
25	CA	\$80,000	M
45	NY	\$150,000	D
55	WA	\$100,500	M
18	TX	\$85,000	S
...	...	...	...



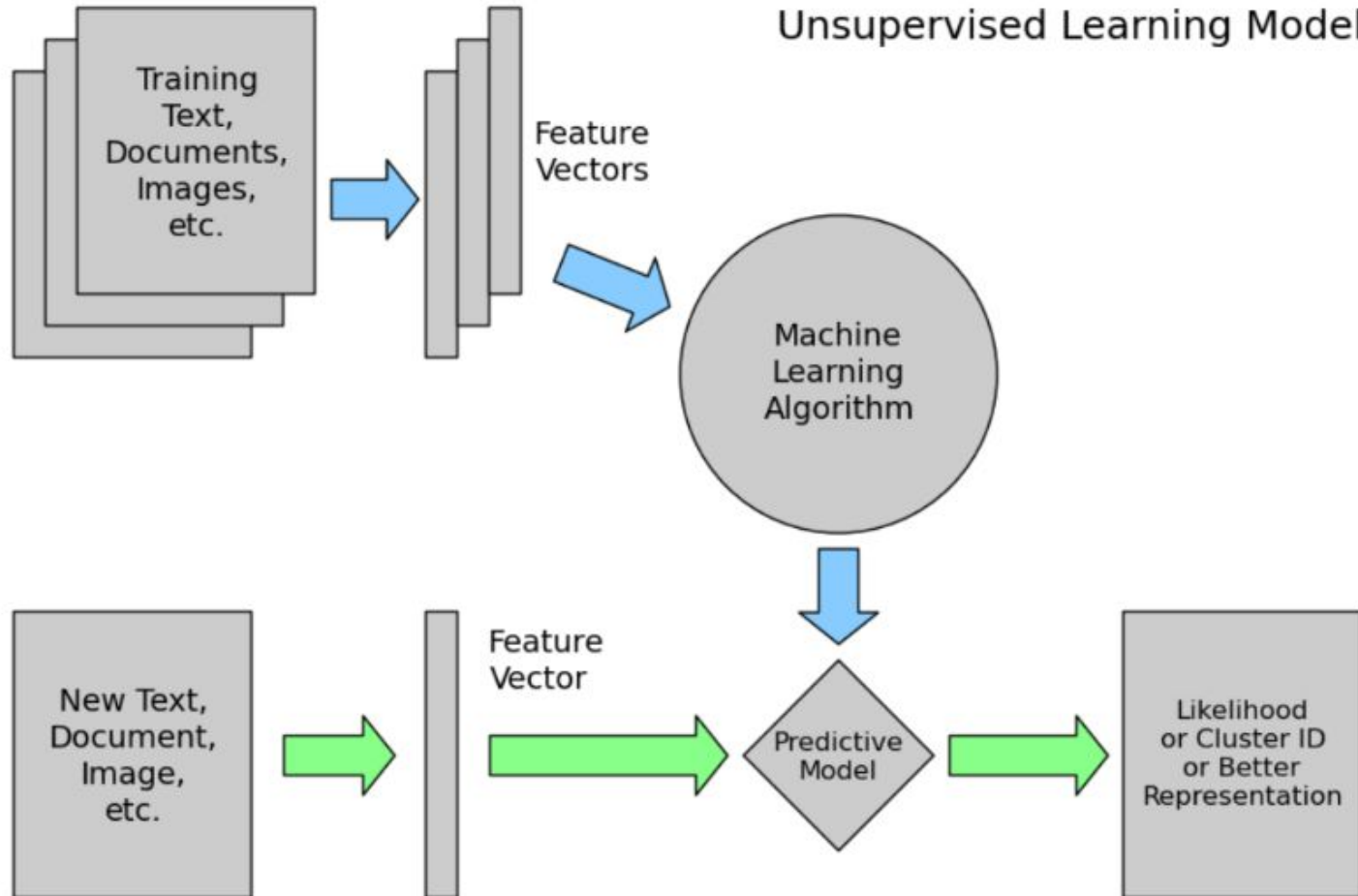
No labels



Naturally occurring
(hidden) structure

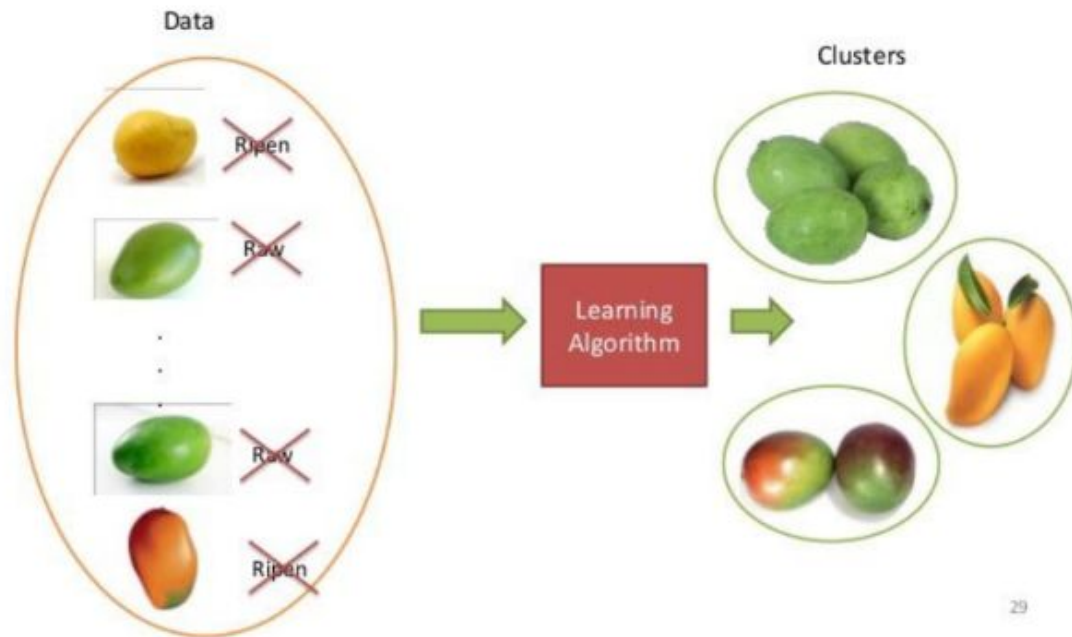
Unsupervised Learning

Unsupervised Learning Model



Unsupervised Learning

Unsupervised Learning



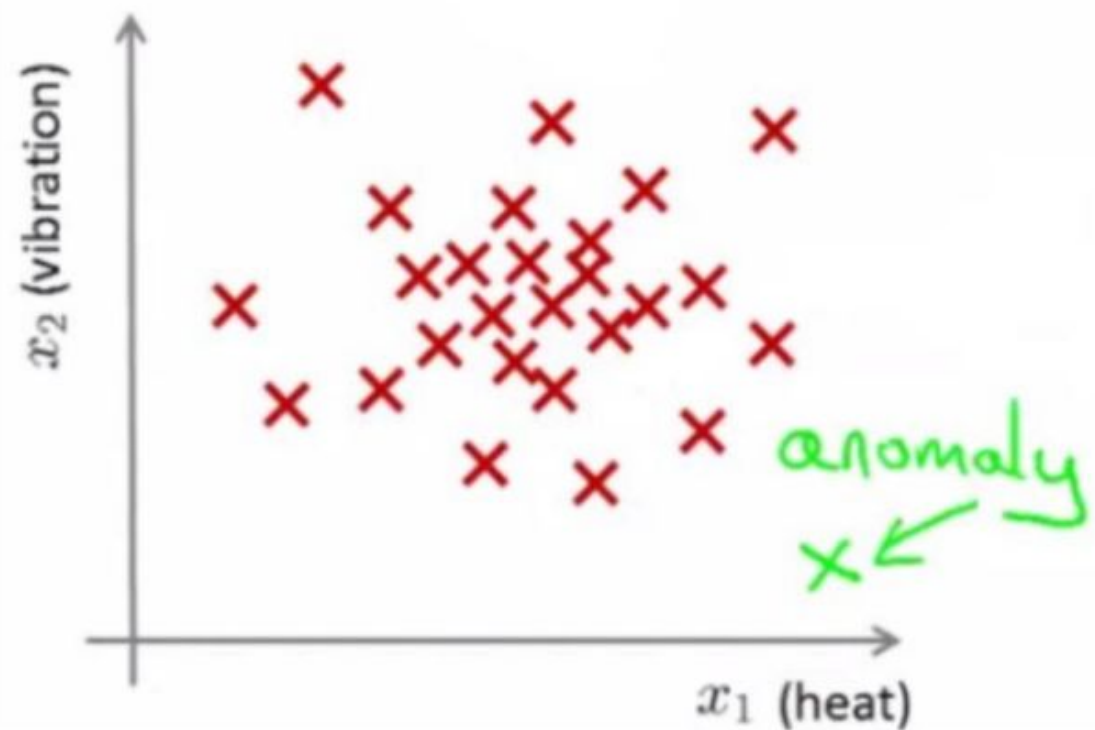
Unsupervised Learning



Objective is simply to divide above Images into N groups
Here ideally $N = 2$

The method can also assume other groups e.g. Group with one object or group with multiple objects

Outlier Detection: identify abnormal patterns



Example: identify engine anomalies

Features:

- Heat generated
- Vibration of engine

Affinity Analysis: identifying frequent item sets

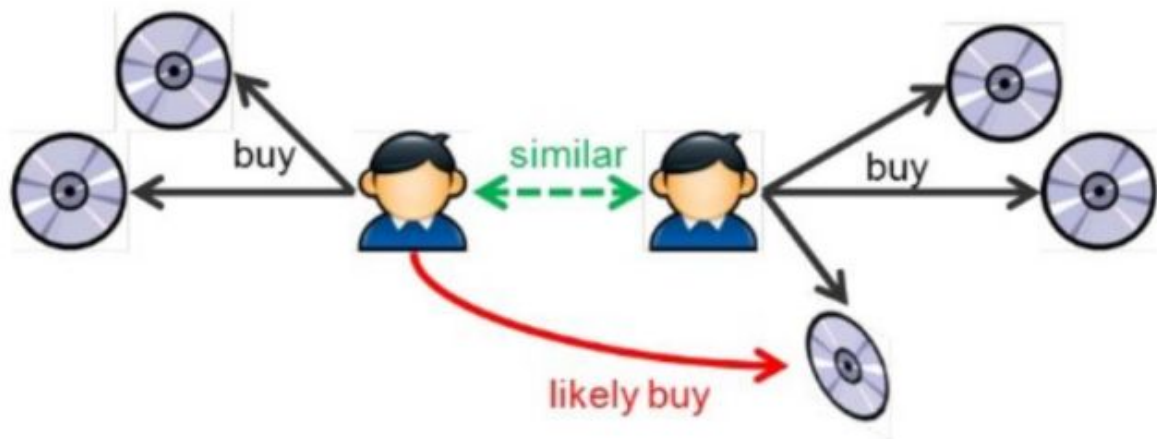
	Item 1	Item 2	Item 3	Item 4	Item 5	...
Tx 1	Y	N	N	Y	N	
Tx 2	Y	N	N	Y	N	
Tx 3	Y	Y	N	Y	N	
Tx 4	N	N	Y	Y	Y	
Tx 5						
...						



	Item 1	Item 2	Item 3	Item 4	Item 5	...
Tx 1	Y	N	N	Y	N	
Tx 2	Y	N	N	Y	N	
Tx 3	Y	Y	N	Y	N	
Tx 4	N	N	Y	Y	Y	
Tx 5						
...						

Goal: identify frequent item set
Techniques: FP Growth, a priori

Product recommendation: predicting “preference”



Collaborative Filtering
Identify users with similar “taste”

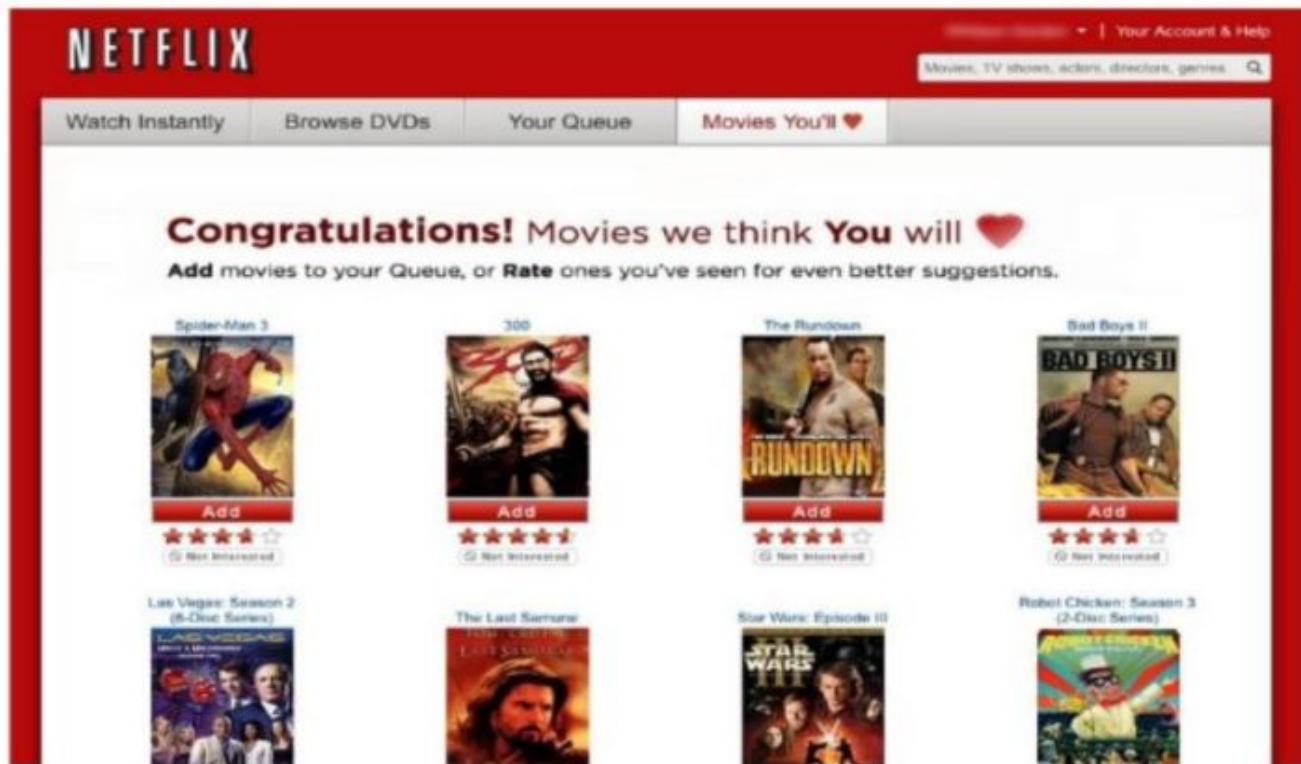
Collaborative filtering -> matrix completion

	Harry potter	X-Men	Hobbit	Argo	Pirates
101	5	2	4	?	?
102	?	?	5	2	?
103	1	2	?	?	3
104					
105					
...					



	Harry potter	X-Men	Hobbit	Argo	Pirates
101	5	2	4	1	3
102	4	1	5	2	3
103	1	2	4	1	3
104					
105					
...					

Example: Netflix



Example: market segmentation



Types of Unsupervised Learning

- **In Clustering** similar instances are grouped, based on their features or properties.
- **Association**: Association rules find associations amongst items within large commercial databases (e.g. **Collaborative filtering**)
 - Discover rules that describe large portions of data, such as people that buy X also tend to buy Y

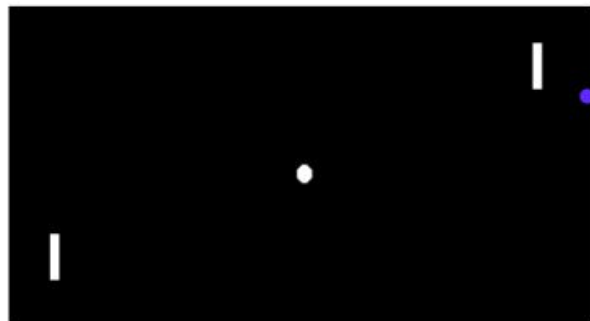
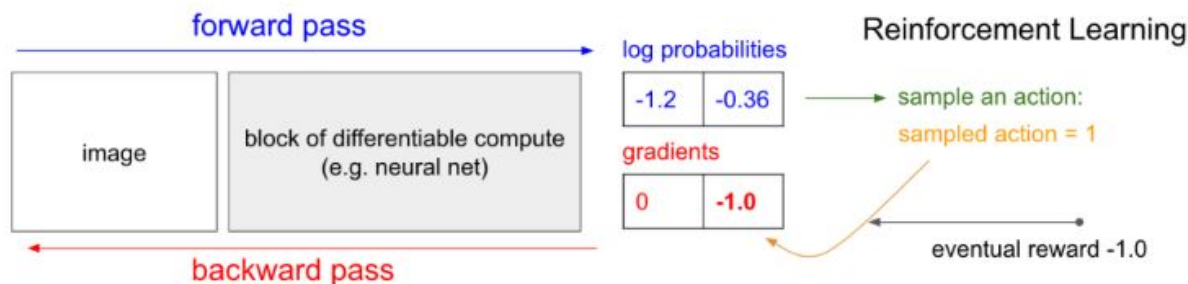
- The similarity between two objects is measured by the **similarity function**
 - The distance between those two object is measured.
 - Shorter the distance higher the similarity, conversely longer the distance higher the dissimilarity.

Reinforcement learning

- Reinforcement learning is a type of dynamic programming that trains algorithms using a system of reward and punishment.
- A reinforcement learning algorithm, or agent, learns by interacting with its environment without intervention from a human by maximizing its reward and minimizing its penalty.
 - The agent receives rewards by performing correctly and penalties for performing incorrectly.
- Reinforcement learning contrasts with other machine learning approaches in that the algorithm is not explicitly told how to perform a task, but works through the problem on its own.

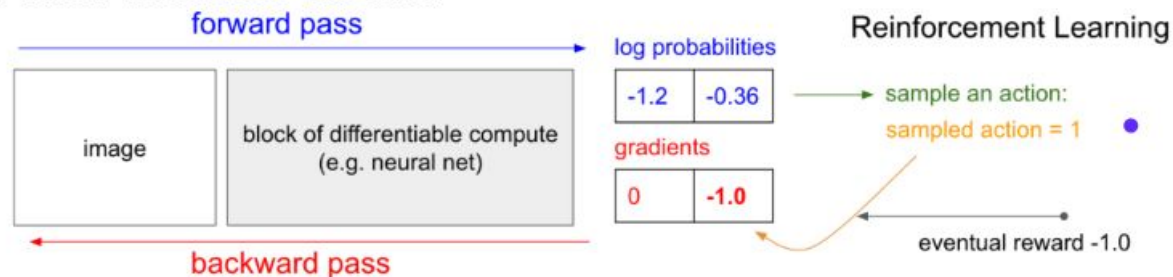
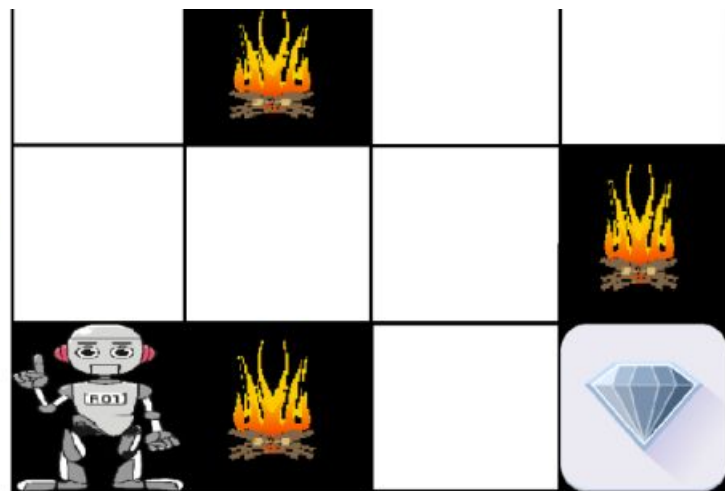
Reinforcement learning

- As an agent, which could be a self-driving car or a program playing chess, interacts with its environment, receives a **reward state** depending on how it performs, such as driving to destination safely or winning a game.
- Conversely, the agent receives a **penalty** for performing incorrectly, such as going off the road or being checkmated.
 - The agent over time makes decisions to **maximize its reward and minimize its penalty** using dynamic programming.
- The advantage of this approach to artificial intelligence is that it allows an AI program to learn without a programmer spelling out how an agent should perform the task.



Reinforcement learning

- The agent is supposed to find the **best possible path to reach the reward**.
- The goal of the robot is to get the reward that is the diamond and avoid the hurdles that is fire.
- The robot learns by trying all the possible paths and then choosing the path which gives him the reward with the least hurdles.
- **Each right step will give the robot a reward and each wrong step will subtract the reward of the robot.**
- The total reward will be calculated when it reaches the final reward that is the diamond.



Classification Workflow

What is a classifier: A classifier is a machine learning model that is used to discriminate different objects based on certain features.

- Whenever you perform classification, the first step is to understand the problem and identify potential features and label.
- Features are those characteristics or attributes which affect the results of the label.
- For example, in the case of a loan distribution, bank manager's identify customer's occupation, income, age, location, previous loan history, transaction history, and credit score.
- These characteristics are known as features which help the model classify customers.